```yaml
---
candidate:
  name: "Robin Macharg"
  title: "Senior / Principal iOS & Multi-Platform Software Engineer"
  phone: "+44 (0)7377 719820"
  email: "robin.macharg@gmail.com"
  linkedin: "https://www.linkedin.com/in/robinmacharg"
  github: "https://github.com/robinmacharg"
  stackoverflow: "http://stackoverflow.com/users/2431627/robin-macharg"
metadata:
  target_roles:
    - "Senior iOS Developer"
  last_updated: "2026-08-03"
semantic_tags:
  candidate_name: "candidate_name"
  candidate_title: "candidate_title"
  candidate_contact: "candidate_contact"
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  role_heading: "role_heading"
  date_range: "date_range"
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  bullet_secondary: "bullet_secondary"
  project_item: "project_item"
  education_item: "education_item"
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  role_requirements_heading: "role_requirements_heading"
  role_requirements_item: "role_requirements_item"
skills_raw:
  status: "NEEDS_UPDATE"
  languages_apis:
    - "Objective-C", "Swift (SwiftUI, Combine, async/await)", "Python (FastAPI, Flask, Pylons)",
"C/C++", "C#", "Kotlin (KMM)", "Java", "Perl", "TypeScript/JavaScript", "SQL", "Shell Scripting"
  ios_mac_ecosystem:
    - "CoreData", "RealmDB", "Alamofire", "ARKit", "Fastlane", "TestFlight", "xcodebuild",
"Cocoapods", "Carthage", "SPM", "RxSwift", "PromiseKit", "XCFrameworks", "Xamarin.iOS"
  gui_web_frameworks:
    - "UIKit", "Winforms", "XAML", "Qt", "PyQt", "OpenGL", "Vue/TypeScript", "React Native",
"Ionic", "HTML/CSS/SASS"
  backend_devops_cloud:
    - "AWS", "Google Cloud (Cloud Run, Firebase, Crashlytics)", "Docker", "Kubernetes",
"PostgreSQL", "MongoDB", "NGinx", "JBoss", "Puppet", "Jenkins", "GitHub Actions", "RESTful APIs"
  graphics_prototyping_hardware:
    - "Figma", "Sketch", "Adobe CC (Photoshop, Illustrator)", "Blender", "3D Printing", "Parametric
CAD (Fusion 360)", "Raspberry Pi", "Robotics/Electronics"
---
```


```{=typst}
#align(center)[
  #set text(18pt, weight: "bold")
  #text("Robin Macharg")
]
#parbreak()
```

Senior / Principal iOS & Multi-Platform Software Engineer

+44 (0)7377 719820 - robin.macharg@gmail.com

[LinkedIn](#) | [GitHub](#) | [StackOverflow](#)

Professional Summary

Principal iOS Engineer *with extensive experience architecting secure, high-performance applications and SDKs using Swift, Objective-C, MVVM, and modern Clean Architecture patterns.*

Security-focused developer *with previous SC clearance, experienced in designing VPN-secured enterprise communications and conducting penetration testing for government and enterprise clients.*

Performance and scale optimizer *who has built and maintained SDKs supporting over 25 million daily users, driving improvements in memory usage, application startup time, and network efficiency.*

Agile delivery champion *and Certified Scrum Master with hands-on experience automating deployment pipelines using Fastlane, GitHub Actions, and custom CI/CD scripting to reduce business risk.*

Security & Performance Highlights

- **Enterprise Security Implementations:** Directed security testing and integrated enterprise-level MDM deployment (AirWatch) for Dyson's professional iOS deployment.
- **High-Concurrency SDK Architecture:** Maintained core video streaming delivery and telemetry SDKs serving 25M+ active users at Sky, integrating Conviva analytics and Freewheel.
- **Deployment Automation:** Re-architected mapping and deployment SDK pipelines, converting manual multi-hour releases to 10-minute automated commands at Living Map.
- **Low-Level Crash Diagnostics:** Managed multi-platform Apple libraries at Bugsnag, resolving complex C++ and Objective-C stack trace and memory crash logs.
- **Academic & Agile Foundations:** Joint BSc (Hons) in Artificial Intelligence & Computer Science, plus Scrum Master (CSM) certification, reinforcing agile delivery best practices.

Education, Certifications & Credentials

- **BSc (Hons) Artificial Intelligence and Computer Science (Joint)** — University of Edinburgh
- **Certified Scrum Master (CSM)** — Scrum Alliance
- **Security Clearance:** Previous SC1 Security Clearance and 6 concurrent DBS (highly relevant to cybersecurity environments)
- **Engineering Management:** Proven leadership experience as Lead Mobile Developer and Head of Mobile Development across agile multi-functional teams

Selected Key Roles

Senior iOS Developer | Sky

October 2021 – April 2022

Keywords: [Swift, Kotlin Multiplatform, SwiftUI, Video Streaming, Analytics, Agile]

- Served as Senior Developer on the Core Video Mobile SDK team supporting Sky, NowTV, Peacock, and Comcast mobile applications.
- Maintained a hybrid Swift and Kotlin Multiplatform Mobile (KMM) SDK and reference apps used daily by over 25 million customers.
- Integrated analytics platforms (Conviva, Adobe) and video streaming delivery technologies (Freewheel, YoSpace, Smooth Stream, DASH) to optimize performance.
- Rewrote the internal reference app in SwiftUI and managed bi-weekly automated release cycles.
- Developed custom Probot/Octokit GitHub tools, auto-assign integrations, and repository hooks to improve team workflows.

Principal iOS Developer | Amiosec Ltd. (Contract)

March 2019 – December 2019

Keywords: [Xamarin, Visual Studio, C#, Objective-C/Swift Interop, EWS, ReactiveUI, Security]

- Served as Senior iOS developer on a custom VPN-secured Exchange client for the UK Government's secure Foxhound project.
- Built high-performance cross-language interoperability between Xamarin/C# and Objective-C/Swift using .NET Embedding.
- Architected and implemented a reactive MVVM pattern utilizing ReactiveUI, DynamicData, Realm.NET, and Polly.
- Reverse-engineered Exchange Web Services (EWS) via SSL Man-in-the-Middle (MiM) proxying using Charles Proxy to verify secure communications.

Senior Apple Engineer | Bugsnag Inc.

January 2020 – June 2020

Keywords: [Objective-C, C++, iOS, macOS, watchOS, tvOS, Open Source, Crash Reporting]

- Senior Cocoa and C++ Developer for multi-platform Apple ecosystems (iOS, macOS, watchOS, tvOS) serving enterprise clients.
- Maintained open-source Bugsnag libraries for Cocoa and C/C++ alongside automated CI testing via Travis and Buildkite.
- Provided specialized 3rd line engineering support handling complex stack traces, crash log analysis, and root cause isolation.
- Managed safe packaging and distribution of SDK components across CocoaPods, Carthage, Manual Install, and Swift Package Manager (SPM).

Principal iOS Developer | Dyson Ltd. (Contract)

June 2017 – January 2020

Keywords: [iOS, Greenfield, Enterprise MDM, ARKit, RealmDB, Swift, Unit Testing]

- Sole developer on Dyson's first B2B sales iPad app ('Hero'), delivering the initial production build from greenfield to field test in 8 weeks (~13k lines).
- Managed end-to-end security integration, including enterprise MDM deployment (AirWatch) and rigorous penetration testing.
- Integrated ARKit, Vuforia/Unity AR, multithreaded RealmDB, PromiseKit, and custom Xcode project templates.
- Achieved >80% UI and Unit Test coverage to ensure long-term stability and code quality.

Senior Developer | Floww Ltd.

May 2023 – September 2023

Keywords: [iOS, SwiftUI, TCA, Redux, GitHub Actions, CI/CD, Mentoring, Shell Scripting]

- Senior iOS developer for a private investment market platform, implementing features using Redux-like Composable Architecture (TCA) on SwiftUI.
- Promoted engineering best practices: introduced componentization, app startup time improvements, compile-time monitoring, and code review templates.
- Maintained robust CI pipelines utilizing GitHub Actions, custom shell scripting, and git push lint hooks.
- Mentored junior engineering team members and assisted in Angular frontend components debugging.

Earlier Career Highlights

1996 – 2023 Positions across Mobile Development, Geospatial SDKs, Telemetry, Network Analytics, and Full-Stack Systems.

- **Lead Mobile Developer** | *This One Ltd.* (SwiftUI, Combine, Async/Await, Python, Flask, Fastlane, GCP, Firebase, ML)
- **Senior Developer** | *Living Map Ltd.* (iOS, Mapping SDK, XCFrameworks, CocoaPods, Carthage, Background Location, UX)
- **Head of Mobile Development** | *Actual Experience Ltd.* (Objective-C, Swift, CoreData, Realm, ICMP, TCP, Jenkins)
- **Principal Mobile Developer** | *Accident Exchange Group* (iOS, CoreData, CoreLocation, MapKit, RestKit)
- **Senior Middleware & DevOps Developer** | *MySpace* (Java, Python, Middleware, JBoss, Drools, Puppet, Maven, Objective-C)
- **Senior Developer** | *RMS Services Ltd.* (Python, PostgreSQL, RESTful APIs, Asset Discovery)
- **Aerodynamics & Mass Properties Engineer** | *Airbus UK Ltd.* (Python, PyQt, Perl, PDL, OpenGL)
- **Server & Embedded Developer** | *Sift Ltd. / The Appliance Studio / Obtree / Circle.com / Glaxo Wellcome* (Linux, Java, Perl, Embedded Appliances, Web)

Open Source & Key Projects

- **Bricksmith:** SourceForge award-winning LEGO CAD software for macOS (ObjC, C) (Link)
- **React Native Input Spinner:** Open-source contributor to interactive mobile UI components (GitHub)
- **Pycel / Excel Formula Parser:** Python-based Excel dependency graph and formula parser (GitHub)
- **Coding Exercises:** Interview algorithms & solutions (GitHub)

Technical Skills & Expertise

- **Core Platforms & Languages:** Swift (SwiftUI, Combine, async/await), Objective-C, Python (FastAPI, Flask), C/C++, Kotlin (KMM), C#, TypeScript, JavaScript, SQL, Shell Scripting (Bash/Zsh)
- **Security & Mobile Architecture:** MVVM, Clean Architecture, Composable Architecture (TCA), Redux, ReactiveUI, Enterprise MDM (AirWatch), VPN Security, Penetration Testing, SSL MiM Proxying (Charles Proxy)
- **Ecosystem & Tools:** iOS SDK, Xcode, TestFlight, App Store Connect, Fastlane, SPM, CocoaPods, Carthage, XCFrameworks, RealmDB, CoreData
- **Cloud & DevOps:** GitHub Actions, Jenkins, Travis, Buildkite, AWS, Google Cloud Run, Firebase, Docker, Kubernetes, PostgreSQL ```